Peter Beigi, PhD

Machine Learning Engineer and Applied AI Researcher

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Applied AI Researcher with expertise in deep learning, time-series modeling, computer vision, and data-driven optimization. Experienced in building end-to-end ML pipelines from data ingestion and feature engineering to model training, validation, and deployment-ready evaluation. Delivered research and production-oriented ML solutions using large-scale transportation and sensor datasets, with strong focus on robust modeling, interpretability, and measurable performance gains.

MACHINE LEARNING EXPERIENCE

Machine Learning Engineer & Applied AI Researcher
Sep 2022 - Present

FHWA Turner-Fairbank Research Center — McLean, VA

- Leading development of AI-based behavioral and motion prediction models for ADAS using deep learning and reinforcement learning frameworks, and integrating AV behavior models into large-scale microsimulation tools (SUMO, VISSIM).
- Developed and validated perception and planning algorithms using LiDAR, radar, sensor fusion, and video data for enhanced AV situational awareness and mixed-traffic simulation.
- Created computer vision pipelines (OpenCV, PyTorch) to extract vehicle and pedestrian trajectories from real-world datasets for motion prediction and safety analysis.

Graduate Researcher
Sep 2021 - Present

The George Washington University — Washington, DC

- Developed multi-agent reinforcement learning (MARL) frameworks for AV coordination and negotiation in dynamic traffic environments.
- Designed and implemented a real-time data streaming platform using Apache Kafka and PyTorch to process 5G-connected video feeds for trajectory prediction and event detection.
- Applied Bayesian optimization and genetic algorithms for driver and AV model calibration.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments.

Machine Learning Research Intern
Sep 2024 - Dec 2024

moment.ai — Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

Data Analyst Intern
May 2025 - Aug 2025

Mead & Hunt — Washington, DC & Columbia, MD

- Built data-driven corridor simulation for WVDOT & BCDOT, and supporting operational scenario testing and optimization studies for smart mobility planning.

Data Analyst Intern
May 2024 - Aug 2024

Arcadis — Atlanta, GA

- Performed geospatial data analysis (MySQL, ArcGIS) and simulation-oriented corridor evaluation (VISSIM) for GDOT projects, translating large datasets into actionable operational and planning insights.

SELECTED PUBLICATIONS

h-index: 6 / citation: 215

"Diffusion Process-Based Model for Network Trajectory Propagation"	IEEE T-ITS (2025)
"Percolation-Based Cloud Computing Framework for Network Reliability Assessment"	IEEE ICMI (2025)
"Motion Modeling in Unstructured Non-Lane-Based Traffic: A Comparative Study of Deterministic, Stochastic, and Multi-Agent Learning Approaches"	IEEE ICRA (under review)
"Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control"	IEEE VTC (under review)
"A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling"	TRR (under review)
"A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance"	TRR (2025)
"Third Generation Simulation Dataset"	TRR (2024)

TECHNICAL SKILLS

Programming: Python, ROS, MySQL, Git

Libraries: PyTorch, TensorFlow, Diffusers, Transformers, Scikit-learn, Keras, OpenCV, Ultralytics, RLlib, Gym

Tools: CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

EDUCATION

The George Washington University
Sep 2021 - Exp: Dec 2026
PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

Sharif University of Technology
Sep 2016 - Feb 2021
Bachelor of Science in Transportation Engineering - GPA: 3.80/4.00

PROFESSIONAL CERTIFICATIONS

Social and Behavioral Research Certification
Issued by: Collaborative Institutional Training Initiative (CITI Program)
Credential ID: 71300833 | Aug 2021 - Aug 2027

RELEVANT COURSEWORK

- UMD: CMSC 422 Machine Learning (audited)
- UMD: CMSC 472 Deep Learning (audited)
- GW: MAE 6292 Robotics Vision and Perception (4/4)
- GW: SEAS 6520 Autonomous Systems and Robotics (audited)
- Stanford: CS231n Deep Learning for Computer Vision (audited)
- GW: CE 6609 Numerical Methods in Engineering (4/4)
- GW: CE 6800 Data Science (4/4)
- GW: ECE 6130 Big Data and Cloud Computing (4/4)
- UMD: CMSC 426 Computer Vision (audited)
- UMD: CMSC 828 Visual Learning and Recognition (audited)

WORK AUTHORIZATION

No sponsorship needed.